

Code-Layout, Readability and Reusability

Date	15 March 2026
Team ID	XXXXXX
Project Name	PocketSmart AI
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

PocketSmart AI is planned as a modular web application with separate UI, API/core-logic, AI integration and data/storage responsibilities. The checklist below records the intended coding standards. Because no final source-code package was provided with these templates, implementation status is marked conservatively.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Partial	Apply one formatter/linting standard across frontend and backend.
2	Proper File Structure	Files and folders are logically organized	Yes	Recommended separation: frontend, backend, AI/service layer, configuration and documentation.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Use names such as budget, category, preferences, recommendations and allocation.
4	Function / Method Names	Functions are descriptively named	Yes	Use action-oriented names such as validateBudget(), generateRecommendations(), compareOptions().
5	Code Comments	Inline and block comments explain logic	Partial	Document non-obvious budget rules, AI prompt construction and integration logic.
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate validation, recommendation orchestration, budget optimization and external API handling.
7	No Redundant Code	Duplicate or unused code is removed	Partial	Keep shared utilities for validation, formatting and API error handling.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Handle invalid inputs, API failures, timeouts and unavailable recommendation sources.

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Input Validation Module	JavaScript / Python	All planning categories	High
2	Budget Optimization Module	Python / Backend	Home Interior, Party, Jewelry	High
3	Recommendation Orchestration	Python / AI service	All recommendation flows	High
4	API Integration Utility	REST / Python	External product/service and AI integrations	Medium
5	Recommendation Comparison UI	React / JavaScript	All categories	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Architecture is modular and can be separated into clear application layers.
Readability	4	Use descriptive naming and consistent formatting; final score depends on delivered source code.
Reusability	4	Budget, validation, recommendation and comparison modules can serve multiple categories.
Documentation / Comments	3	Document API contracts, configuration, AI prompts and non-obvious business rules.
Overall Score	4	Good target for maintainable implementation; final assessment should be updated after code review.